



Conservation of Mass — Practice Problems

Grade 8 Science • Chemical Change & the Mass of Substances

Grade _____ Class _____ No. _____ Name _____

Part 1 What happens to the mass? 5 patterns Picture each experiment and choose

1. Copper powder on a combustion dish was heated **in open air**. What happens to the mass? Answer
- A : Increases B : Decreases C : Stays the same
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2. Copper was placed in a round-bottom flask, **sealed with a rubber stopper**, and heated. What happens to the mass? Answer
- A : Increases B : Decreases C : Stays the same
-
3. Dilute sulfuric acid was mixed with barium hydroxide solution, and a **white precipitate** (barium sulfate) formed. What happens to the mass? Answer
- A : Increases B : Decreases C : Stays the same
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4. Inside a **sealed jar**, hydrochloric acid was added to sodium bicarbonate, and carbon dioxide was produced. What happens to the mass? Answer
- A : Increases B : Decreases C : Stays the same
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5. Inside an **open beaker (no lid)**, hydrochloric acid was added to sodium bicarbonate, and carbon dioxide was produced. What happens to the mass? Answer
- A : Increases B : Decreases C : Stays the same
-

Part 2 Reasons & mechanism Think in terms of atoms!

6. Which is the most correct reason why the **Law of Conservation of Mass** holds? Answer
- A : The combination of atoms changes, but the kinds and numbers of atoms do not change
- B : New atoms are created and old atoms disappear
- C : The atoms gradually get smaller during the chemical change
- D : No gas is produced in a chemical change
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7. In question 1, what is the **reason the mass increased** when copper was heated in open air? Answer
- A : The number of copper atoms increased B : It got heavier by the amount of heat that was added
C : Oxygen from the air bonded with the copper
D : The copper atoms themselves turned into heavier atoms
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8. In question 5, what is the **reason the mass decreased** in the open beaker (no lid)? Answer
- A : The carbon dioxide disappeared into nothing
B : The carbon dioxide that was produced escaped into the air
C : All of the hydrochloric acid evaporated D : The number of atoms decreased in the reaction
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9. After the experiment in question 4 (the sealed jar), if you **open the lid** and leave it for a while, what happens to the total mass? Answer
- A : Increases B : Decreases C : Stays the same
D : Increases at first, then returns to the original
-

Part 3 Try the calculations The total is the same before and after!

10. When **10 g** of dilute sulfuric acid was mixed with **17 g** of barium hydroxide solution, **23 g** of barium sulfate and some water formed. How many grams of water formed? Answer
- A : 2 g B : 4 g C : 6 g D : 13 g
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11. Sodium bicarbonate and hydrochloric acid were reacted inside a sealed container. The total before the reaction was **58.0 g**. After the reaction, the lid was opened and after a while it became **57.4 g**. How many grams of carbon dioxide escaped? Answer
- A : 0.4 g B : 0.6 g C : 1.6 g D : 57.4 g
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12. **2.0 g** of limestone was put into **10.0 g** of dilute hydrochloric acid (no lid). Gas was produced, and the total after the reaction was **11.2 g**. How many grams of gas were produced? Answer
- A : 0.8 g B : 1.2 g C : 2.0 g D : 8.0 g
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Part 4 True or False Watch for tricks!

13. "If you dissolve 60 g of sugar in 200 g of water, the total mass will be 260 g even after the sugar dissolves and can no longer be seen." Answer
- : ○ True × : × False
-

14. "In a reaction where gas escapes, the Law of Conservation of Mass does not hold."

Answer

☐ : ☐ True ☒ : ☒ False

Part 5 **Explain it**  Write your idea in your own words

15. Even though it is the same reaction (sodium bicarbonate + hydrochloric acid), the mass stayed the same in **the sealed jar** (question 4) but decreased in **the open beaker with no lid** (question 5). Explain why this "difference" occurs.



Words to use:

carbon dioxide

the air

leaves

trap
